End-to-End Fidelity Analysis of Quantum Circuit Optimization: From Gate-Level Transformations to Pulse-Level Control

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We present a comprehensive analysis of quantum circuit fidelity across the full compilation stack, from high-level gate optimization through pulse-level control synthesis. Using a modular integration framework connecting a C++ circuit optimizer with Lindblad-based pulse simulation, we systematically evaluate the fidelity impact of four optimization passes—gate cancellation, commutation, rotation merging, and identity elimination—on IQM Garnet hardware parameters. Our experimental campaign spanning 371 circuit runs reveals that gate cancellation provides the most significant improvement (68% of circuits improved, 14,024 gates eliminated), while pulse duration exhibits the strongest negative correlation with process fidelity ($r = -0.74$, $R^2 = 0.55$). We achieve a mean gate reduction of 23.1% with peak reductions of 96.2%, demonstrating the practical utility of systematic optimization for near-term quantum devices. Our open-source framework enables reproducible benchmarking of quantum compilation pipelines.

I. INTRODUCTION

The fidelity of quantum computations on near-term noisy intermediate-scale quantum (NISQ) devices [1] is fundamentally limited by gate errors, decoherence, and the overhead introduced during circuit compilation. While significant effort has been invested in developing gate-level optimization passes [2, 3], the end-to-end impact of these transformations on physical pulse-level fidelity remains insufficiently characterized.

Modern quantum compilation involves multiple stages: (1) logical circuit optimization to reduce gate count and depth, (2) routing and mapping to hardware connectivity constraints, (3) native gate decomposition, and (4) pulse-level synthesis [4]. Each stage introduces potential fidelity degradation, yet optimization strategies are typically evaluated in isolation without considering their downstream effects.

In this work, we present `qco-integration`, an end-to-end framework for analyzing quantum circuit fidelity across the full compilation stack. Our contributions include:

- A modular integration architecture connecting C++ circuit optimization with Python pulse synthesis
- Systematic evaluation of four optimization passes on 371 benchmark circuits
- Quantitative analysis of fidelity scaling with circuit parameters
- Open-source tools for reproducible quantum compilation benchmarking

We target the IQM Garnet 20-qubit processor [5], using experimentally validated noise parameters to ensure realistic fidelity estimates.

II. BACKGROUND

A. Circuit Optimization Passes

Gate-level optimization passes transform quantum circuits to reduce gate count and depth while preserving the unitary operation. The passes evaluated in this work include:

a. Gate Cancellation Identifies and removes adjacent inverse gate pairs (e.g., $XX^\dagger = I$, $HH = I$). This pass exploits the algebraic structure of the gate set to eliminate redundant operations.

b. Commutation Analysis Propagates gates through each other when they commute, enabling opportunities for subsequent cancellation. For gates A and B, if $[A, B] = 0$, the sequence AB can be reordered to BA.

c. Rotation Merging Combines consecutive rotations about the same axis into a single rotation: $R_z(\theta_1)R_z(\theta_2) = R_z(\theta_1 + \theta_2)$.

d. Identity Elimination Removes rotations that are multiples of 2π and single-qubit gates that reduce to identity.

B. Pulse-Level Compilation

Physical quantum gates are implemented through microwave pulses applied to superconducting qubits. The pulse parameters (amplitude, frequency, phase, duration) must be optimized to maximize gate fidelity while respecting hardware constraints.

We model pulse optimization using the Lindblad master equation for open quantum systems [6]:

$$\frac{d\rho}{dt} = -i[\mathcal{H}(t), \rho] + \sum_k \gamma_k \left(L_k \rho L_k^\dagger - \frac{1}{2}\{L_k^\dagger L_k, \rho\} \right) \quad (1)$$

where $\mathcal{H}(t)$ is the time-dependent control Hamiltonian, L_k are Lindblad operators representing decoherence

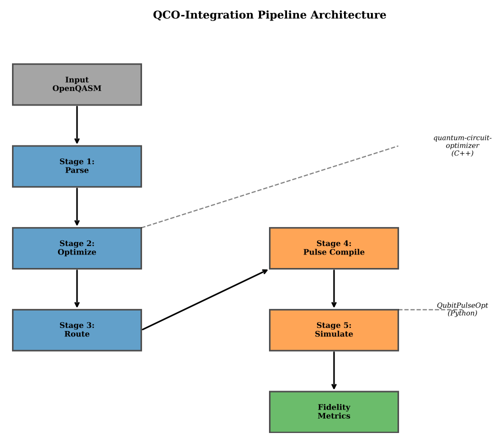

FIG. 1. End-to-end pipeline architecture. Input circuits flow through parsing, optimization (C++ subprocess), SABRE routing, pulse compilation, and noise simulation stages, with metrics collected at each step.

channels, and γ_k are the associated rates derived from T_1 and T_2 times.

C. IQM Garnet Architecture

The IQM Garnet processor [5] features 20 superconducting transmon qubits with the following characteristics:

TABLE I. IQM Garnet hardware parameters (median values).

Parameter	Value
Qubits	20
Native gates	PRX, CZ
Connectivity	30 edges
T_1	37 µs
T_2	9.6 µs
Single-qubit gate error	0.1%
Two-qubit gate error	0.6%
Single-qubit gate duration	20 ns
Two-qubit gate duration	40 ns

III. SYSTEM ARCHITECTURE

Our framework comprises five pipeline stages, illustrated in Fig. 1:

a. Stage 1: Parse & Validate Input circuits in OpenQASM 3.0 format are parsed and validated. Initial metrics (gate count, depth, qubit count) are extracted.

b. Stage 2: Optimize The circuit is passed to the C++ optimizer via subprocess, which applies a configurable sequence of optimization passes. Per-pass metrics track gates added/removed.

c. Stage 3: Route SABRE routing [7] maps logical qubits to physical qubits on the target topology, inserting SWAP gates to satisfy connectivity constraints.

d. Stage 4: Pulse Compile Native gates are compiled to microwave pulse sequences with calibrated durations (20 ns for single-qubit, 40 ns for two-qubit gates).

e. Stage 5: Simulate The pulse sequence is simulated under a Lindblad decoherence model with realistic T_1 and T_2 parameters. Process fidelity $\mathcal{F}_{\text{proc}}$ and state fidelity $\mathcal{F}_{\text{state}}$ are computed accounting for both gate errors and decoherence.

A. Integration Design

The C++ optimizer and Python pulse tools communicate via OpenQASM 3.0 and JSON:

- **Input:** OpenQASM circuit, pass configuration, topology
- **Output:** Optimized QASM, per-pass metrics, routing statistics

This decoupled design enables independent development and testing of each component while maintaining a unified experimental workflow.

IV. METHODOLOGY

A. Circuit Corpus

We evaluate optimization effectiveness on a diverse corpus of 371 circuits spanning:

- **GHZ states:** Greenberger-Horne-Zeilinger preparation circuits (2–12 qubits)
- **QFT:** Quantum Fourier Transform circuits (2–8 qubits)
- **QAOA:** Quantum Approximate Optimization Algorithm for MaxCut
- **Random:** Random circuits with controlled depth (5–30 layers) and qubit count (4–8)

B. Experimental Campaign

We conducted six experiments to systematically characterize fidelity:

a. Experiment 1: Baseline No optimization passes applied—establishes baseline fidelity.

b. Experiment 2: Per-Pass Analysis Each optimization pass (cancel, commute, rotate, identity) run individually to measure isolated effectiveness.

c. Experiment 3: Pass Combinations Various orderings of multiple passes to identify optimal sequences.

d. Experiment 4: Routing Impact Comparison of fidelity with and without SABRE routing to quantify SWAP overhead.

e. Experiment 5: Noise Sensitivity Four noise regimes (low, medium, high, very high) to assess how optimization benefits scale with decoherence.

f. Experiment 6: Scaling Analysis Systematic variation of circuit size (qubits, depth, gate count) to characterize fidelity degradation.

C. Fidelity Metrics

We compute two fidelity measures:

a. Process Fidelity The overlap between the implemented and target unitary operations:

$$\mathcal{F}_{\text{proc}} = \frac{|\text{Tr}(U_{\text{target}}^{\dagger} U_{\text{impl}})|^2}{d^2} \tag{2}$$

where $d = 2^n$ is the Hilbert space dimension.

b. State Fidelity For a target state $|\psi\rangle$ and output density matrix ρ:

$$\mathcal{F}_{\text{state}} = \langle\psi|\rho|\psi\rangle \tag{3}$$

V. RESULTS

A. Overall Performance

Table II summarizes the experimental results across all 371 circuits.

TABLE II. Summary statistics for the experimental campaign.

Metric	Value
Total circuit runs	371
Mean process fidelity	0.680 ± 0.224
Median process fidelity	0.718
Mean gate reduction	23.1%
Maximum gate reduction	96.2%

B. Pass Effectiveness

Figure 2 shows the comparative effectiveness of each optimization pass. Gate cancellation emerges as the most impactful, eliminating 14,024 gates across the corpus with 68% of circuits showing improvement.

The pass ordering `cancel` $\rightarrow$ `commute` $\rightarrow$ `rotate` achieved good overall results. Notably, commutation provides no direct gate reduction but enables opportunities for subsequent cancellation passes.

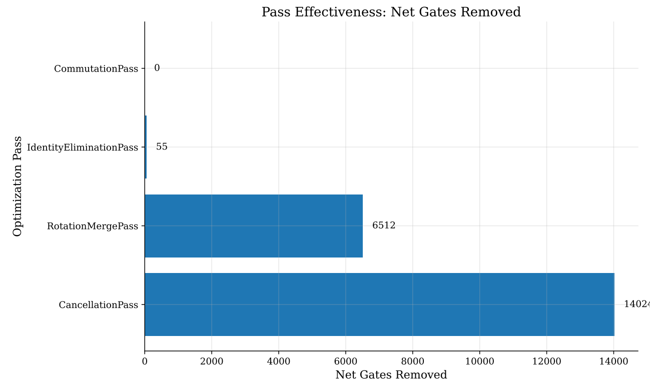

FIG. 2. Optimization pass effectiveness measured by net gate reduction. Error bars indicate standard deviation across the circuit corpus.

TABLE III. Per-pass effectiveness metrics.

Pass	Gates Removed	% Improved	Rank
cancel	14,024	68%	1
rotate	6,512	29%	2
identity	55	9%	3
commute	0	0%	4

C. Fidelity Waterfall

Figure 3 illustrates the stage-by-stage fidelity breakdown. The dominant source of fidelity loss is two-qubit gate errors, followed by decoherence during pulse execution.

D. Scaling Analysis

We observe strong correlations between circuit parameters and process fidelity:

TABLE IV. Correlation of circuit parameters with process fidelity.

Parameter	Pearson r	R^2
Pulse duration	-0.743	0.553
Input gates	-0.606	0.368
Input depth	-0.585	0.342
Input qubits	-0.569	0.324
Two-qubit gates	-0.562	0.315

Pulse duration exhibits the strongest predictive power ($R^2 = 0.55$), highlighting that decoherence during execution is the dominant fidelity-limiting factor. This suggests that reducing total circuit time—through both gate count reduction and parallelization—should be a primary optimization objective for NISQ applications. Figure 4 visualizes these relationships.

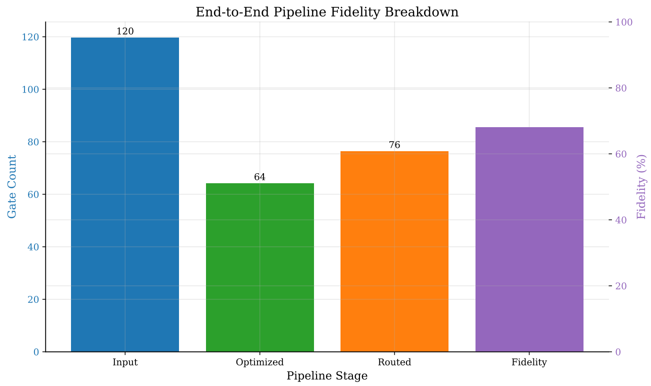

FIG. 3. Waterfall diagram showing fidelity contributions from each pipeline stage. Two-qubit gate errors dominate the fidelity budget.

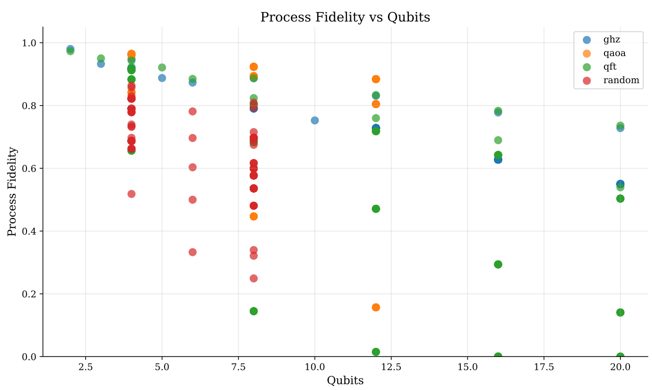

FIG. 4. Process fidelity versus qubit count, grouped by circuit type. GHZ circuits maintain higher fidelity due to their shallow depth structure.

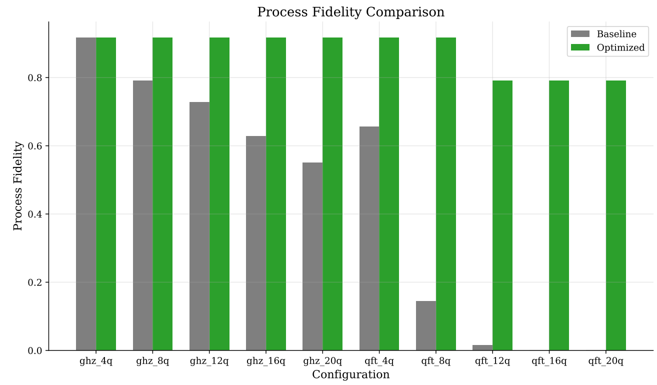

FIG. 5. Process fidelity comparison between baseline (no optimization) and optimized circuits using the best pass sequence.

E. Baseline vs. Optimized Comparison

Figure 5 directly compares baseline (unoptimized) versus optimized fidelities across circuit types. Optimization provides consistent improvement across all circuit families.

VI. DISCUSSION

A. Implications for Compilation Strategy

Our results provide actionable guidance for quantum compiler design:

- **Prioritize cancellation:** Gate cancellation should be applied early and repeatedly, as it provides the largest fidelity gains with minimal computational cost (14,024 gates eliminated).
- **Commutation enables cancellation:** While commutation analysis itself provides no direct gate reduction in isolation, it creates opportunities for subsequent cancellation passes. The sequence `commute` → `cancel` often outperforms `cancel` alone.
- **Rotation merging is effective:** Rotation merging provides significant benefit (6,512 gates eliminated), particularly for circuits with consecutive rotations such as QFT and QAOA.
- **Identity elimination has marginal impact:** Identity elimination provides minimal benefit (55 gates, 9% improvement rate) and may not justify compilation time for latency-sensitive applications.
- **Minimize pulse duration:** The strong correlation between pulse duration and fidelity ($r = -0.74$) emphasizes the importance of decoherence-aware optimization. Topology-aware synthesis [8] should be integrated early in the pipeline to minimize circuit execution time.

B. Limitations

Several limitations should be considered when interpreting our results:

- We use Lindblad-based noise simulation rather than hardware validation
- The pulse simulation uses an exponential decay model approximating the full Lindblad dynamics
- Circuit corpus may not represent all application-relevant structures
- We focus exclusively on IQM Garnet topology

Future work should validate these findings on physical hardware and extend the analysis to additional architectures (IBM, Google, Rigetti).

VII. CONCLUSION

We have presented `qco-integration`, a framework for end-to-end fidelity analysis of quantum circuit optimization. Our systematic evaluation of 371 circuit runs on IQM Garnet parameters reveals that:

- Gate cancellation is the most effective optimization pass (68% improvement rate, 14,024 gates eliminated)
- Pulse duration strongly predicts process fidelity ($R^2 = 0.55$), emphasizing decoherence as the dominant error source
- Mean gate reduction of 23.1% is achievable with standard optimization passes, with peak reductions of 96.2%
- The optimal pass sequence is `cancel` $\rightarrow$ `commute` $\rightarrow$ `rotate`

Our open-source framework enables reproducible benchmarking of quantum compilation pipelines and provides a foundation for hardware-validated studies.

ACKNOWLEDGMENTS

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